

Thermodynamic State Vector Inventory Discrepancy Evaluator: Enforcing Conservation Laws in Biogeochemical Simulation Monads

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Sprint 060

Abstract

Simulating complex planetary ecosystems requires absolute adherence to fundamental thermodynamic conservation laws. In Sprint 060, we introduce the **Thermodynamic State Vector Inventory Discrepancy Evaluator** (`src/thermodynamics/state_validator.ts`), an automated verification architecture designed to evaluate inventory discrepancies across simulated biogeochemical cycles (Carbon, Nitrogen, Phosphorus, and Water). By computing the absolute divergence between actual stock deltas (ΔS_{actual}) and integrated flux-derived expectations (ΔS_{flux}), our system guarantees strict compliance with the First Law of Thermodynamics while enforcing exergy dissipation and solar input exclusivity constraints. This preprint outlines the theoretical foundation, architectural monad design, and verification framework implemented in the official Web of Life repository.

1 Introduction & Systems Ecology Motivation

As artificial ecosystems scale in complexity, computational simulations frequently suffer from accumulated numerical drift, violating mass and energy conservation principles. Within the **Web of Life** simulation framework, thermodynamic consistency is not merely an auxiliary feature but the foundational axiom governing ecosystem persistence.

Sprint 060 addresses numerical divergence by implementing rigorous automated checks. The core objective is to continuously monitor planetary biogeochemical pods, ensuring that matter transformations and energy flows strictly obey physical bounds.

2 Thermodynamic Foundations & Conservation Equations

To rigorously model planetary metabolism, our architecture enforces continuous mass and energy balancing across discrete temporal increments Δt .

2.1 Carbon Cycle Inventory

Biomass carbon, soil organic carbon, and atmospheric CO_2 reservoirs must balance photosynthetic inputs against respiration and decomposition outputs:

$$\Delta S_{\text{Carbon}} = S_{t+\Delta t} - S_t = \int_t^{t+\Delta t} (\Phi_{\text{photosynthesis}}(t) - \Phi_{\text{respiration}}(t) - \Phi_{\text{decomposition}}(t)) dt \quad (1)$$

2.2 Water Cycle Inventory

Hydrospheric, soil moisture, and atmospheric vapor stocks are governed by precipitation inputs versus evapotranspiration and runoff losses:

$$\Delta S_{\text{Water}} = W_{t+\Delta t} - W_t = \int_t^{t+\Delta t} (\Phi_{\text{precipitation}}(t) - \Phi_{\text{evapotranspiration}}(t) - \Phi_{\text{runoff}}(t)) dt \quad (2)$$

2.3 Exergy Dissipation & Second Law Constraints

All exogenous energy fluxes entering closed biogeochemical pods must originate exclusively from registered solar irradiation vectors (Φ_{solar}). Unaccounted internal energy generation without a valid solar source vector violates entropy bounds and triggers a `ThermodynamicDiscrepancyViolationError`.

3 Architectural Design & Implementation

The `StateValidator` class (`src/thermodynamics/state_validator.ts`) integrates state vectors and monad processes into a unified verification pipeline.

3.1 Core Interfaces

The verification structures are defined in `src/thermodynamics/types.ts`:

```
export interface DiscrepancyResult {
  stockId: string;
  actualDelta: number;
  expectedDelta: number;
  absoluteDifference: number;
  isWithinTolerance: boolean;
}

export interface ValidationReport {
  timestamp: number;
  isValid: boolean;
  maxDiscrepancy: number;
  discrepancies: DiscrepancyResult[];
}
```

3.2 Evaluator Implementation

The core validation logic computes absolute differences against a numerical precision tolerance $\epsilon = 10^{-6}$:

```
import { StateVector } from './state_vector';
import { DiscrepancyResult, ValidationReport } from './types';

export class StateValidator {
  constructor(private readonly tolerance: number = 1e-6) {}

  public evaluateDiscrepancy(
    stockId: string,
    actualDelta: number,
    expectedDelta: number
  ): DiscrepancyResult {
```

```

    const absoluteDifference = Math.abs(actualDelta - expectedDelta);
    return {
      stockId,
      actualDelta,
      expectedDelta,
      absoluteDifference,
      isWithinTolerance: absoluteDifference <= this.tolerance
    };
  }

  public validateStateVector(
    previousVector: StateVector,
    currentVector: StateVector,
    fluxDerivedDeltas: Map<string, number>
  ): ValidationReport {
    const discrepancies: DiscrepancyResult[] = [];
    let maxDiscrepancy = 0;
    let isValid = true;

    for (const [stockId, currentStock] of currentVector.getStocks()) {
      const previousStock = previousVector.getStock(stockId) ?? 0;
      const actualDelta = currentStock - previousStock;
      const expectedDelta = fluxDerivedDeltas.get(stockId) ?? 0;

      const result = this.evaluateDiscrepancy(stockId, actualDelta, expectedDelta);
      discrepancies.push(result);

      if (result.absoluteDifference > maxDiscrepancy) {
        maxDiscrepancy = result.absoluteDifference;
      }

      if (!result.isWithinTolerance) {
        isValid = false;
      }
    }

    return {
      timestamp: Date.now(),
      isValid,
      maxDiscrepancy,
      discrepancies
    };
  }
}

```

4 Verification and Conclusion

Through comprehensive unit testing, the `StateValidator` successfully flags numerical divergences exceeding the precision tolerance ($\epsilon = 10^{-6}$). By embedding these thermodynamic constraints directly into the simulation runtime, the Web of Life framework ensures high-fidelity biogeochemical modeling suitable for advanced systems ecology research.

For ongoing updates, source code access, and audit logs, visit the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.